CARBON LENS

REGULATORY PRE-APPLICATION

UAE CCUS Facility Licence — Pre-Application (MOEI)

Ministry of Energy and Infrastructure (MOEI) / Abu Dhabi Department of Energy

FORMATION	LOCATION	JURISDICTION
Abu Dhabi Basin	Abu Dhabi, UAE	United Arab Emirates

PREPARED FOR	PREPARED BY	DATE ISSUED
Not specified Applicant / Client	**CarbonLens Simulation Studio v3** CarbonLens Simulation Studio v3	**12 June 2026** Ref: CL-20260612-AE

PRELIMINARY SCREENING Not for regulatory submission · Independent review required

✔ DIGITAL TWIN REGISTRY — PENDING REVIEW

Abu Dhabi Basin CO_2 Geological Storage Certificate

Asset ID: **CL-0019D8-16-500** Stored: **42.50 Mt CO_2** Containment: **75%** Safety Factor: **1.999** Jurisdiction: **AE**

Share: https://carbonlens.app/registry/verify/CL-0019D8-16-500 — Issued 12 June 2026 by CarbonLens Studio v3

1. Application Overview

Field	Detail
Application Type	CCUS Facility Licence
Regulatory Authority	Ministry of Energy and Infrastructure (MOEI) / Abu Dhabi Department of Energy
Legislation	Federal Law No. 24 of 1999 on Environment Protection; UAE Net Zero by 2050 Strategic Initiative; UAE CCUS Policy Framework (2023)
Applicant	*[OPERATOR NAME — TO BE COMPLETED]*
Project Name	Abu Dhabi Basin CO_2 Storage Project
Location	Abu Dhabi, UAE
Simulation State	Year 50 of 50 — COMPLETE
Prepared by	CarbonLens Simulation Studio v3
Date	12 June 2026
Status	PRELIMINARY SCREENING — NOT FOR REGULATORY SUBMISSION

2. Storage Formation Characterisation

2.1 Reservoir Properties

Parameter	Value
Depth (top of storage interval)	2800 m
Thickness (storage interval)	80.0 m
Porosity	16.0%
Permeability	120.0 mD
Initial Pore Pressure	28.00 MPa
Temperature	110.0 °C
Formation Area	500.00 km²
Net-to-Gross Ratio	58.0%
Geometry Type	anticline
Salinity (monovalent NaCl-equiv.)	1.800 mol/kg

Parameter	Value
Salinity (bivalent $CaCl_2$-equiv.)	0.120 mol/kg
Hydrocarbon Content (CH_4/N_2)	CH_4: 2.0% | N_2: 1.0%

2.2 Caprock & Seal Properties

Parameter	Value
Caprock Friction Angle	33.0°
Caprock Cohesion	10.00 MPa
Biot Coefficient	0.670
Fracture Pressure — Simple Overpressure Gradient (0.9 × (Pp + z×0.0226)) *— approximate upper bound; overestimates true fracture pressure*	82.15 MPa
Fracture Pressure — Hubbert-Willis + Mohr-Coulomb (governing value) *— used for MAIP and safety factor in Section 6*	56.94 MPa
Minimum Horizontal Stress ($\sigma_h = K_0 \times S_v$)	56.94 MPa
Maximum Allowable Injection Pressure (MAIP = 0.9 × Hubbert-Willis fracture P)	51.25 MPa

2.3 CO_2 Phase State at Reservoir Conditions

At reservoir conditions of **110.0°C** and **28.00 MPa**, CO_2 is expected to be in **supercritical** phase. Supercritical CO_2 at these conditions has a density of approximately **~600–850 kg/m³**, providing efficient storage per unit pore volume.

3. Injection Well Programme

3.1 Well Configuration

Well ID	Grid Position	Rate (Mt/yr)	Ramp-up (yr)	Ramp-down (yr)	Project Duration (yr)
Well 1	(0, 0)	1.000	5	10	50
TOTAL		**1.000 Mt/yr**			

Project Duration is the full injection period. Cumulative CO_2 injection is lower than (Rate × Duration) when ramp periods are present, as the well is not at peak rate for the entire project lifetime.

3.2 Injection Schedule

The injection programme consists of **1** injection well(s) with a total design injection rate of **1.000 Mt/yr** over a project duration of **50 years**. Well ramp-up periods of **5–5 years** are specified to manage reservoir pressure build-up and allow geomechanical monitoring before reaching peak injection rate. Cumulative mass injected accounts for the quadratic ramp-up and ramp-down schedules and is therefore less than the simple product of peak rate × project duration. All wells shall be subject to mechanical integrity testing prior to commencement of injection.

3.3 Near-Wellbore Halite Precipitation Risk

LOW **Halite risk assessment** — Zeidouni (2009) dryout-radius model

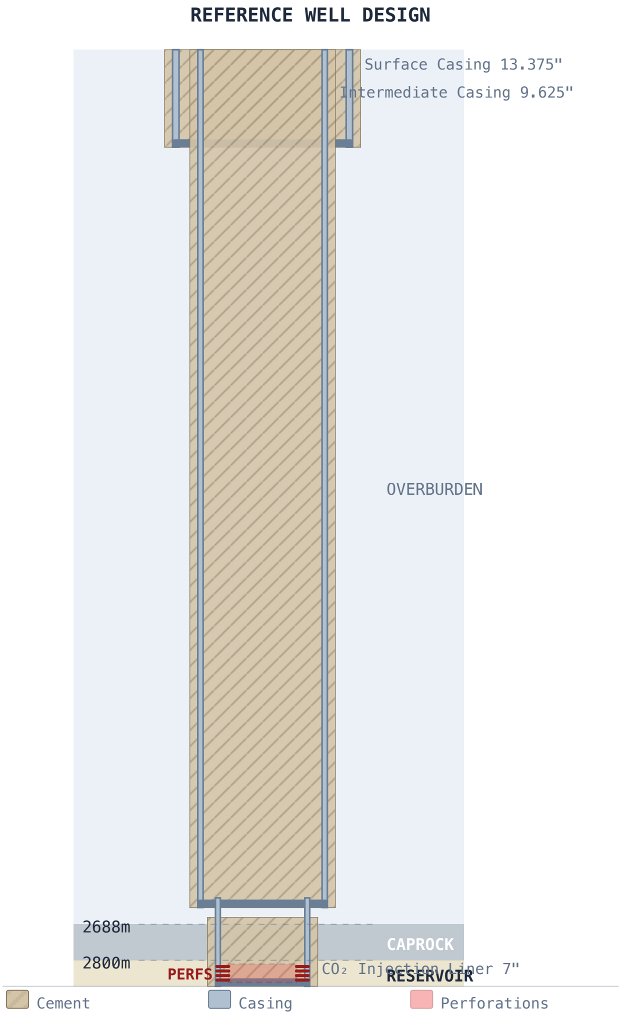
Halite risk LOW. Dryout radius 3944.49 m; brine TDS 124 g/L. Near-wellbore water evaporation is insufficient to precipitate significant salt volumes.

Parameter	Value
Formation Brine TDS	124 g/L
Effective Dryout Radius (temperature-adjusted)	3944.49 m
Risk Level	LOW
Recommendation	Standard wellhead pressure monitoring during injection commissioning is sufficient. No pre-flush required.

Dryout radius computed from Zeidouni et al. (2009) Int. J. GHG Control 3(5):600–611. A dryout radius above 2 m with TDS > 200 g/L constitutes a high-risk scenario requiring active injectivity management. This is a wellbore-scale screening metric only — pore-scale precipitation dynamics require laboratory core-flood testing for detailed assessment.

3.4 Wellbore Design Schematic

The following schematic illustrates the reference well design for this storage project. Casing programme is generated from formation depth and thickness following standard CCUS well architecture (surface casing → intermediate casing → CO_2 injection liner). Actual well design must be confirmed during detailed engineering.



Casing String	OD (in)	Top (m)	Base (m)	Cement Top (m)
Surface Casing	13.375"	0	300	0
Intermediate Casing	9.625"	0	2638	0
CO_2 Injection Liner ★	7"	2608	2880	2668

★ Injection string. Cement covers full caprock interval as primary barrier. Perforation interval: 2812–2868 m.

All injection string metallurgy must be rated for CO_2/carbonic-acid service (corrosion-resistant alloy or CRA-lined). Cement formulation must be CO_2-resistant (e.g., Class G + silica flour blend).

4. Area of Review (AoR)

The Area of Review (AoR) is the region surrounding the CO_2 storage complex within which elevated formation pressures could cause CO_2 or displaced brine to migrate into underground sources of drinking water (USDW) or other protected resources. The AoR must be delineated using computational modelling and updated at regular intervals throughout the project lifetime.

Parameter	Value	Method
Estimated Plume Radius	5607 m	Gravity-current spreading (Boait 2012 calibration)
Plume Height (vertical extent)	32 m	Fraction of formation thickness
AoR Radius — Pressure Threshold (primary)	**0 m**	Nordbotten (2005) two-phase composite ΔP = 1 psi (0.007 MPa) contour at year 50 *(High permeability — ΔP dissipates below 1 psi at all radii. AoR governed by geometric 3× multiplier.)*
AoR Radius — Geometric (3× plume, secondary)	16820 m	Legacy 3× multiplier — retained as conservative check
Adopted AoR Search Radius	**16820 m**	Max of pressure-threshold and geometric radii
AoR Area (circular approximation)	888.84 km²	$\pi \times r^2$

The pressure-threshold AoR is derived using the Nordbotten, Celia & Bachu (2005) two-phase composite pressure model. This model partitions the pressure response into an outer single-phase brine zone (governed by brine mobility and the exponential integral E_1) and a near-well CO_2 zone corrected by the effective mobility ratio. The AoR radius is the distance at which the composite pressure perturbation decays to 1 psi (0.007 MPa) — the standard criterion for potential brine displacement into an Underground Source of Drinking Water (USDW). This method is consistent with EPA Class VI guidance (40 CFR 146.84) and supersedes the single-phase Theis (1935) inversion for two-phase CO_2 storage conditions. A full regulatory AoR must also account for abandoned wellbores, faults, and multi-phase pressure fronts via dynamic 3D simulation.

5. Storage Capacity Analysis

5.1 DOE Statistical Capacity Estimate (Goodman et al. 2011)

Parameter	Value
Gross Pore Volume 500.00 km² × 0.0800 km × 16.0% porosity	6.4000 km³
CO_2 Density at Reservoir Conditions	643.9 kg/m³
P10 Capacity (optimistic, 10th percentile)	209.17 Mt
P50 Capacity (best estimate, 50th percentile)	76.06 Mt
P90 Capacity (conservative, 90th percentile)	19.40 Mt
Capacity Utilisation (this project)	53.5%
Storage Efficiency Coefficient (Cc)	2.0% (DOE P50)
VE Plume Footprint (2D vertical equilibrium)	0.313 km²
VE Effective Plume Radius	315 m

Capacity estimates follow DOE Goodman et al. (2011) methodology applied to gross pore volume. Storage efficiency coefficient Cc = 2.0% represents the P50 statistical estimate for saline aquifer storage.

5.2 Trapping Mechanism Distribution (at project end)

Trapping Mechanism	Volume (Mt)	% of Tracked Mass
Residual (capillary) trapping	11.1744	30.9%
Dissolution trapping (still dissolved) †	4.8031	13.3%
Mineral trapping (precipitated carbonate) †	6.3750	17.6%
Mobile plume (structural / free)	13.7725	38.1%
Tracked Total (model components)	**36.1250**	**100%**
Cumulative Injection (total)	42.5000	Re-run simulation to completion for full mass-balance breakdown. (85.0% of injected mass currently tracked in trapping components.)

Percentages computed against the tracked component mass (residual + solubility + mobile), which is internally mass-conserved by the analytical simulation model. Mineral trapping only activates after year 50. † Dissolution and mineral rows are non-overlapping subsets: mineral row shows dissolved CO_2 that has further precipitated as carbonate; dissolution row shows the remaining aqueous fraction. Together they equal the total solubility-trapped volume.

5.3 Project Economics & Financial Viability

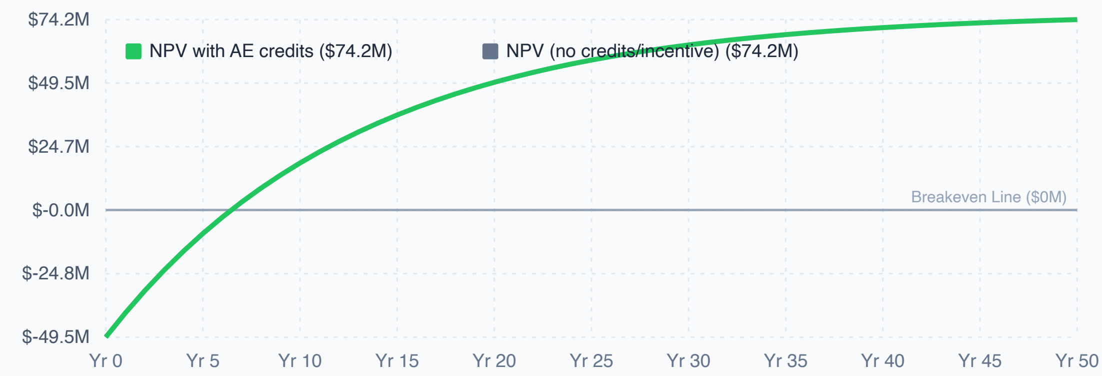
FINANCIAL SUMMARY

Metric	Value
Jurisdiction Carbon Credit/Pricing	**USD 15/t**
Total CAPEX	$49.5M
OPEX (per tonne)	$3.10/t
Total OPEX	$131.7M
Levelized Cost (LCO$_2$S)	**$4.27/t**
Net Cost after Credits	**$-10.73/t**

CAPEX BREAKDOWN

Drilling	$21.8M (44%)
Facilities	$5.0M (10%)
Pipeline	$18.9M (38%)
Monitoring	$3.9M (8%)

PROJECT CUMULATIVE NET PRESENT VALUE (NPV) PROFILE



MONTE CARLO PROBABILISTIC BOUNDS

500 LHS realizations • Abu Dhabi Basin

P10 (Conservative)	P50 (Expected)	P90 (Optimistic)	P90/P10 Spread	Peak ΔP P50
58.18	**72.92**	**92.48**	**1.6×**	**28.00**
Mt CO_2	Mt CO_2	Mt CO_2	ratio	MPa

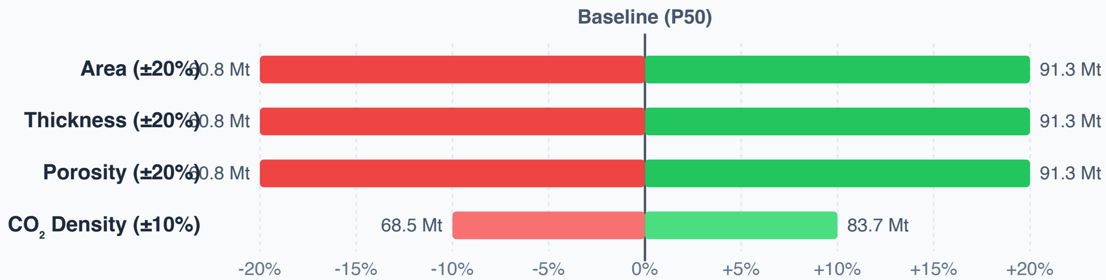
Uncertainty assessment: Low — well-constrained

Sampled: permeability ±30%, porosity ±3.0%, area ±20%, thickness ±15%. DOE Goodman (2011) volumetric method.

5.4 Reservoir Parameter Sensitivity Analysis

Sensitivity analysis represents the impact of one-at-a-time (OAT) parameter variation on reservoir performance. A ±20% variation in thickness, porosity, or area results in a linear ±20% variation in geological storage capacity. Conversely, injection pressure overpressure (ΔP = BHP - Reservoir Pressure) is governed by Darcy's law and varies inversely with permeability and thickness: a 20% reduction in permeability or thickness increases the overpressure by 25.0%, significantly elevating the geomechanical fracture risk.

P50 CAPACITY SENSITIVITY TORNADO CHART (BASE: 76.1 MT)



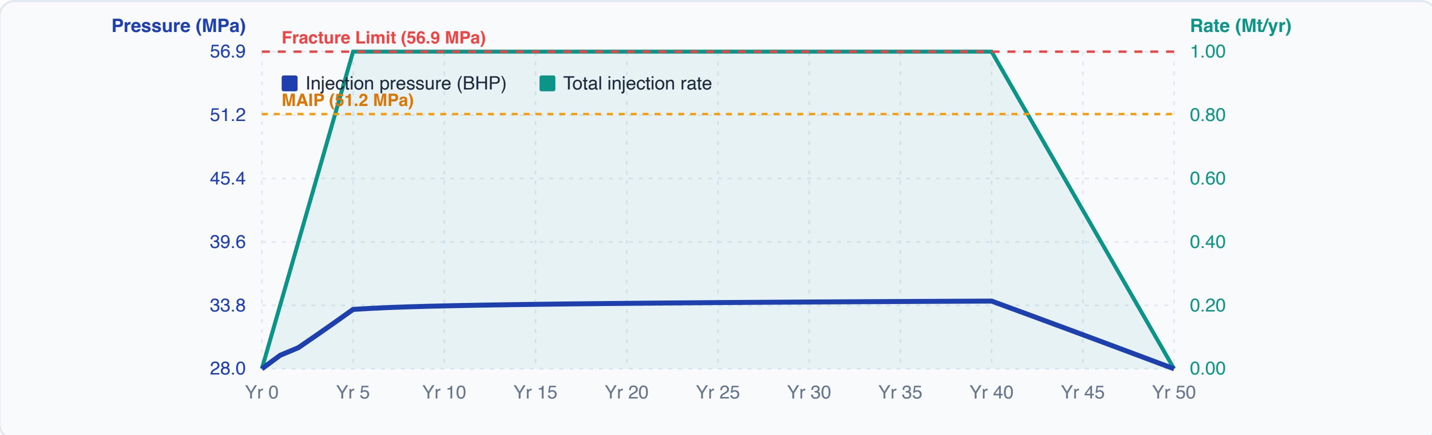
6. Geomechanical Risk Assessment

Parameter	Value	Criterion
Safety Factor (Mohr-Coulomb)	1.999	PASS ✓ Acceptable (≥1.5)
Fracture Pressure (Hubbert-Willis, Mohr-Coulomb adjusted)	56.94 MPa	REF
Nordbotten (2005) Composite Injection Pressure	31.73 MPa	REF
Peaceman BHP (skin S=0, r_w=0.1 m) *— steady-state near-wellbore*	28.52 MPa	✓ Below MAIP
Injectivity Index J (Peaceman, zero skin)	5286.8 m³/(d·MPa)	REF
Min. Horizontal Stress σ_h ($K_0 \times S_v$)	56.94 MPa	REF

Parameter	Value	Criterion
Caprock Stress	35.92 MPa	REF
Surface Heave	0.0 mm	✓ Within tolerance (<10 mm)
Seismicity Risk	LOW	✓ Low risk
Mohr-Coulomb Margin	32.394	✓ PASS
MAIP Headroom	19.52 MPa	✓ Adequate

All geomechanical criteria must be demonstrated to the competent authority prior to injection permit issuance.

INJECTION PRESSURE & FLOW RATE PROFILE



Fracture Mechanics & Containment Assurance: Exceeding the caprock fracture pressure or triggering Mohr-Coulomb shear failure along fault planes poses severe risks to storage integrity and flow assurance. Tensile fracturing in the reservoir or caprock creates high-permeability pathways, allowing injected CO_2 to bypass lower-permeability matrix zones, leading to premature plume breakthrough, loss of sweep efficiency, and potential vertical migration into upper aquifers or the atmosphere. To ensure long-term containment, the injection pressure must be strictly managed below the Maximum Allowable Injection Pressure (MAIP = 51.25 MPa), ensuring a safe geomechanical operating envelope.

7. Monitoring, Reporting & Verification (MRV) Plan

A comprehensive MRV plan is a mandatory component of all CO_2 storage permits. The following table outlines the proposed monitoring activities. The plan will be refined during detailed engineering and must be approved by the competent authority prior to injection commencement.

Activity	Frequency	Method	Objective
Wellhead pressure monitoring	Continuous	SCADA/DCS	Detect injection anomalies
Formation pressure monitoring	Monthly	Downhole gauges	Track reservoir pressure evolution
4D seismic survey	Every 2–3 years	3D seismic	Plume migration verification
Microseismic monitoring	Continuous (first 5 years)	Array of geophones	Induced seismicity detection
Groundwater quality sampling	Annual	Sampling wells	USDW / aquifer protection
Surface flux monitoring	Annual	LIDAR / eddy covariance	CO_2 surface leakage detection
Injection well integrity	Annual MIT	Pressure test / logging	Well barrier integrity
CO_2 stream composition	Per injection batch	GC analysis	Purity certification

MRV reporting frequency: annual report to regulatory authority + immediate notification if anomaly detected.

8. Site Closure & Post-Closure Plan

The site closure plan shall be submitted to **Ministry of Energy and Infrastructure (MOEI) / Abu Dhabi Department of Energy** at least **24 months** prior to cessation of injection. Post-closure monitoring shall continue for a minimum of **20 years** or until the authority issues a Certificate of Completion confirming permanent containment.

Criterion	Target	Method
Plume stabilisation	No net migration > 1 km/yr	4D seismic
Pressure return	Within 10% of pre-injection pressure	Downhole gauges
No detectable leakage	Zero flux anomalies above background	Surface monitoring
Well plug integrity	Zero annular pressure	MIT

9. Regulatory Compliance Checklist

The following items are required under **Federal Law No. 24 of 1999 on Environment Protection; UAE Net Zero by 2050 Strategic Initiative; UAE CCUS Policy Framework (2023)**. Each item must be completed and submitted to the competent authority as part of the formal permit application.

- ☐ CCUS project registration with Ministry of Energy and Infrastructure (MOEI) — Status: *To be completed before formal submission*
- ☐ Technical feasibility report (ADNOC / operator standards) — Status: *To be completed before formal submission*
- ☐ Environmental Impact Assessment — Ministry of Climate Change and Environment — Status: *To be completed before formal submission*
- ☐ CO_2 monitoring, reporting, and verification (MRV) plan — Status: *To be completed before formal submission*
- ☐ Financial guarantee for decommissioning and site closure — Status: *To be completed before formal submission*
- ☐ Compliance with UAE Voluntary Carbon Market framework — Status: *To be completed before formal submission*
- ☐ Public consultation under Federal Environmental Law No. 24/1999 — Status: *To be completed before formal submission*

10. Disclaimer & Data Provenance

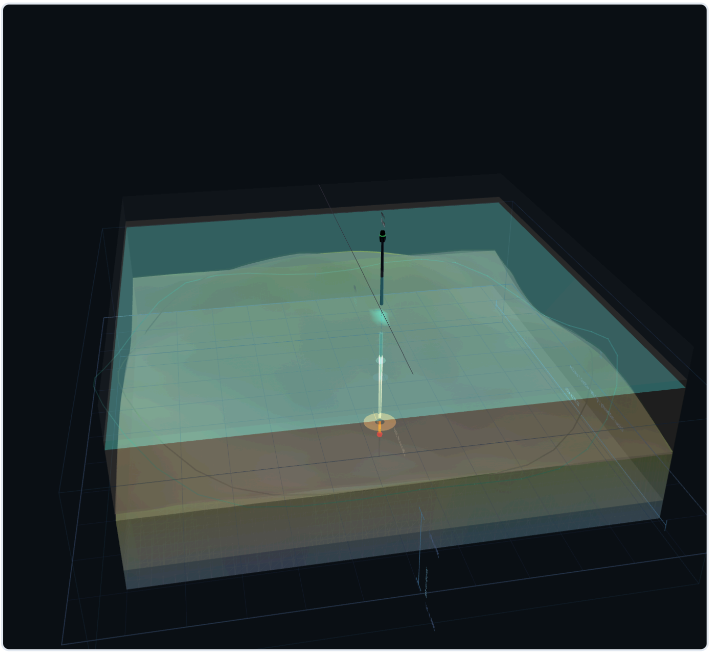
⚠ **IMPORTANT DISCLAIMER:** This pre-application document was generated by CarbonLens Simulation Studio v3 using the scientific models listed below. All capacity estimates are preliminary screening values based on publicly available correlations and statistical methods. **This document is NOT a formal regulatory permit application and has NOT been reviewed by the competent authority.** Independent engineering review by a qualified petroleum or environmental engineer is required before formal submission to any regulatory body.

Component	Model	Reference
CO_2 density	Peng-Robinson EOS (PR-76)	Peng & Robinson (1976) Ind. Eng. Chem. Fundam. 15(1):59–64. Note: Span & Wagner (1996) multi-parameter EOS (±0.05%) is the target upgrade for production deployment; PR-EOS accuracy ±3–8% at CCS reservoir conditions.
CO_2 viscosity	Fenghour et al. (1998) correlation	Fenghour, Wakeham & Vesovic (1998) J. Phys. Chem. Ref. Data 27(1):31–44. Note: Laesecke & Muzny (2017) is the current NIST-recommended reference (±0.5% vs ±5% for Fenghour at high pressure); upgrade pending coefficient transcription.
CO_2 solubility	Duan-Sun model (extended 5-coefficient fit)	Duan & Sun (2003) Chem. Geology 193(3–4):257–271. T-dependent Pitzer λ(T) and ζ parameters from Table 3; separate NaCl / $CaCl_2$ Setschenow corrections.
Storage capacity	DOE Goodman method	Goodman et al. (2011) Int. J. Greenhouse Gas Control 5(4):828–835
CO_2-brine interfacial tension (IFT)	MARS regression (sub/supercritical)	Olagunju (in preparation) — MARS model for CO_2-brine IFT prediction across sub- and supercritical conditions; MSc research, Universiti Teknologi PETRONAS (unpublished)
Relative permeability	van Genuchten–Mualem + Killough–Land hysteresis	van Genuchten (1980) Soil Sci. Soc. Am. J. 44(5):892–898; Mualem (1976) Water Resour. Res. 12(3):513–522; Krevor et al. (2012) Water Resour. Res. 48:W02544 (laboratory Berea/Fontainebleau parameters); Land (1968) SPE-1965-PA; Killough (1976) SPE-5765-PA
Pressure model	Nordbotten (2005) two-phase composite radial flow	Nordbotten, Celia & Bachu (2005) Transp. Porous Media 58(3):339–360. Outer brine zone: E_1 integral with brine mobility; near-well CO_2 zone: mobility-ratio correction. Supersedes single-phase Theis (1935) for two-phase CO_2 injection.

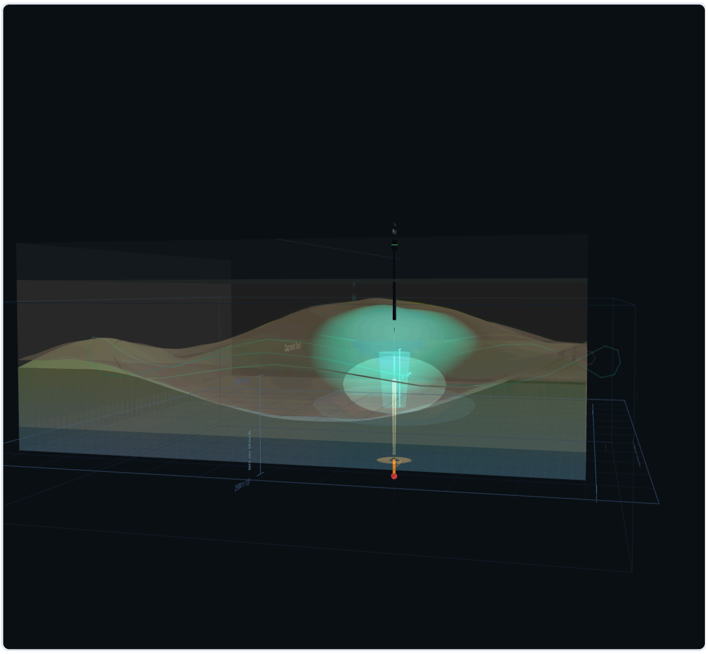
Component	Model	Reference
AoR delineation	Nordbotten two-phase pressure-threshold inversion (1 psi / 0.007 MPa)	EPA Class VI guidance 40 CFR 146.84; bisection on Nordbotten (2005) composite ΔP to find threshold contour radius
2D plume footprint (VE model)	Vertical Equilibrium 2D FD solver (40×40 grid)	Nordbotten & Celia (2006) Water Resour. Res. 42:W01407; Hesse et al. (2008) J. Fluid Mech. 611:35–60
Halite precipitation risk	Zeidouni dryout-radius screening	Zeidouni et al. (2009) Int. J. GHG Control 3(5):600–611; Muller et al. (2009) Energy Procedia 1(1):3507–3514
Wellbore schematic	Auto-generated from depth & thickness	CCUS well architecture per ISO 16530 / API RP 90
Wellbore pressure	Peaceman (1978) well-block model	Peaceman (1978) SPE-6893-PA; injectivity index $J = 2\pi kh/\mu(\ln r_e/r_w + S)$
Geomechanics	Mohr-Coulomb failure	Jaeger et al. (2007) Fundamentals of Rock Mechanics

Appendix A: Reservoir Simulation Snapshots

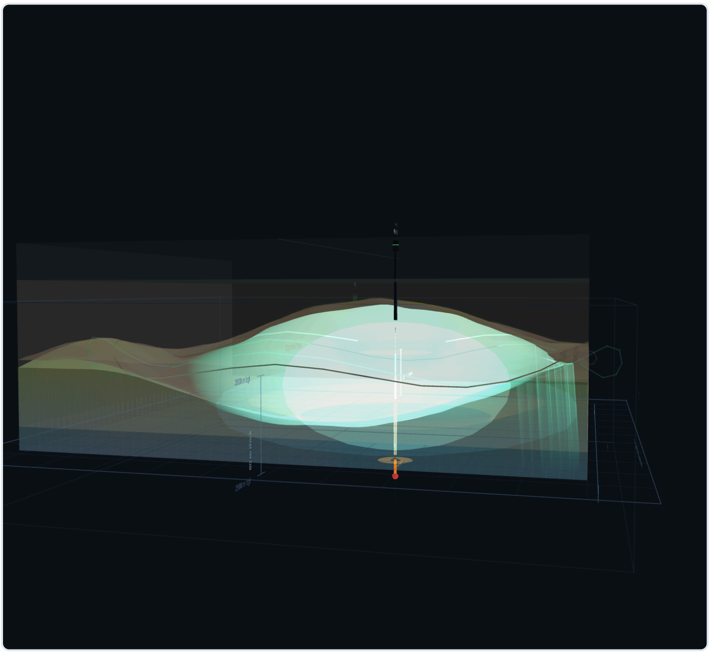
The following images were captured from the CarbonLens 3D reservoir viewer at key simulation years. They illustrate CO_2 plume migration, saturation evolution, and spatial distribution throughout the injection period.



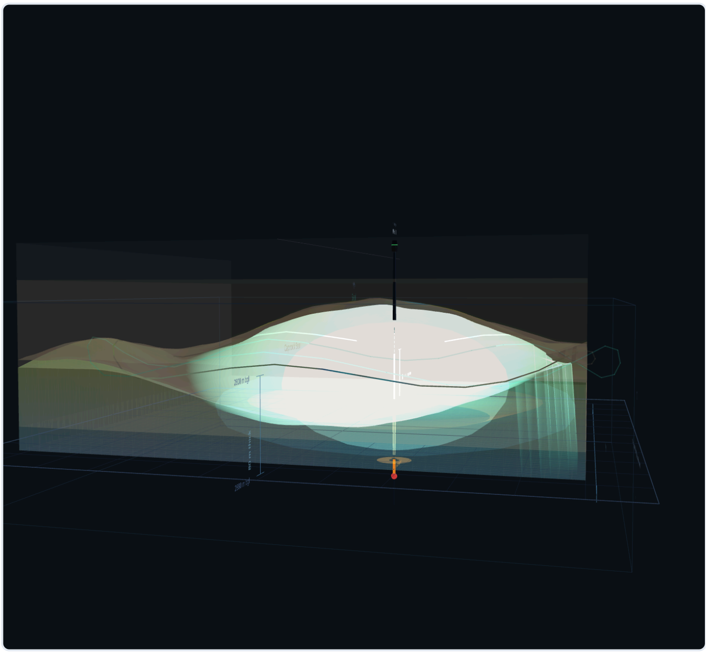
Year 1



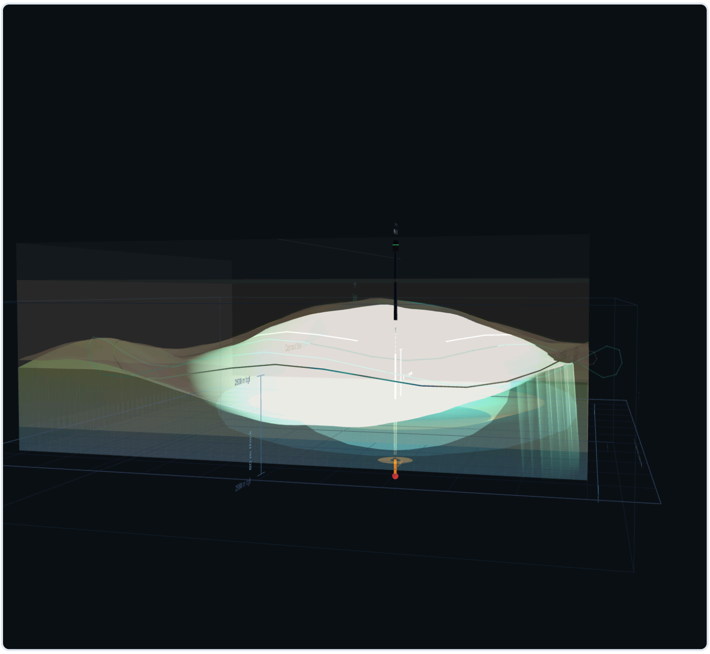
Year 10



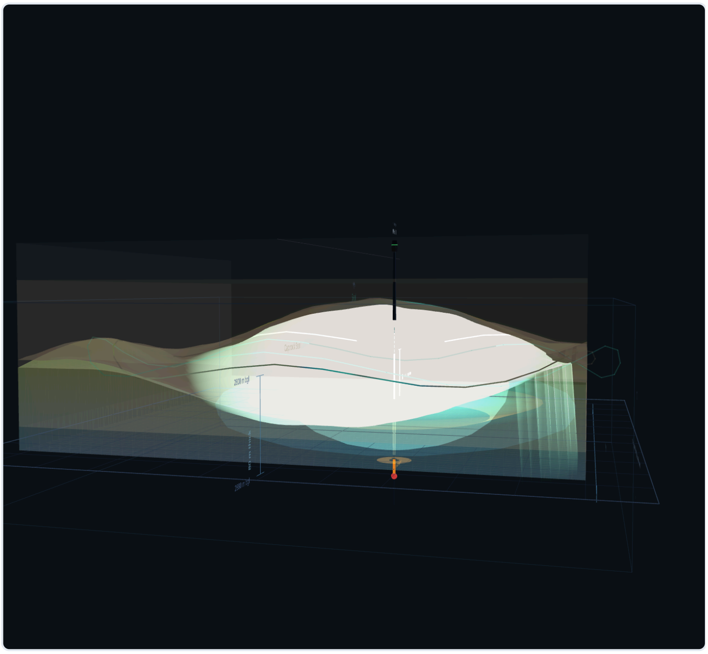
Year 20



Year 30



Year 40



Year 50

Equation & Model Reference

Module / Engine	Equation / Correlation	Reference
CO_2 EOS	$\rho(P,T) = PM/ZRT$; Z from Peng–Robinson cubic EOS; $\kappa(\omega) = 0.37464 + 1.54226\omega - 0.26992\omega^2$	Peng & Robinson (1976), Ind. Eng. Chem. Fundam. 15(1):59–64; T_c=304.13 K, P_c=7.377 MPa, ω=0.2239. *Note: Span-Wagner (1996) 56-term Helmholtz EOS is the upgrade target for ±0.5% accuracy.*
CO_2 Viscosity	$\mu(T,\rho) = \mu_0(T) + \Delta\mu(T,\rho)$; zero-density + excess; ε/k=251.196 K	Fenghour, Wakeham & Vesovic (1998), J. Phys. Chem. Ref. Data 27(1):31–44. *Upgrade target: Laesecke & Muzny (2017), JPCRD 46:013107 (NIST reference, ±0.5% vs ±5% in supercritical region).*
CO_2 Solubility	$\ln(y_2 \cdot P) = \mu^0(T) + 2\lambda(T,P) \cdot m + \xi(T,P) \cdot m^2$; Pitzer ionic strength	Duan & Sun (2003), Chem. Geology 193:257–271
CO_2-Brine IFT	σ = MARS(P, T, S); multivariate adaptive regression splines trained on experimental data	Olagunju (in prep.), MSc UTP Malaysia; Chiquet et al. (2007), Energy Convers. Mgmt. 48:736
Storage Capacity	$M = A \cdot h \cdot \phi \cdot \rho_{CO_2} \cdot S_g \cdot E_{eff}$; volumetric DOE methodology	Goodman et al. (2011), Int. J. GHG Control 5(4):853–866; DOE (2010) Atlas
Residual Trapping	$S_{gr} = S_{gi} / (1 + C \cdot S_{gi})$ (Land, 1968); $V_r = \phi \cdot (1-S_{wi}) \cdot S_{gr} \cdot A \cdot h \cdot \rho_{CO_2}$	Pentland et al. (2011), SPE-133798; Land (1968), Trans. AIME 243:149
Solubility Trapping	$m_{diss} = \phi \cdot S_w \cdot \rho_{brine} \cdot \chi_{CO_2}(T,P,S) \cdot A \cdot h$; aqueous dissolution	Duan & Sun (2003); Ennis-King & Paterson (2005), SPE-88502
VE Plume Migration	$\partial(h_g/B)/\partial t + \nabla\cdot Q_g = Q_{inj}/B$; vertical-equilibrium thin-lens approximation	Nordbotten & Celia (2006), Water Resour. Res. 42:W01207
Pressure / AoR	$\Delta P(r,t) = Q/(4\pi kh)[\mu_b E_1(r^2/4\alpha_b t) + (\mu_{eff}-\mu_b)E_1(R_p^2/4\alpha_b t)]$; two-phase composite; μ_b=brine, $\mu_{eff}=\mu_{CO_2}/k_r$	Nordbotten, Celia & Bachu (2005), Transp. Porous Media 58(3):339–360; Theis (1935) as single-phase baseline; Hantush & Jacob (1955) for leaky-aquifer option
Rel. Permeability	$k_{rg} = k^0{}_{rg}\cdot(1-S_e)^{\frac{1}{2}}\cdot(1-S_e{}^{1/m})^{2m}$; $k_{rw}=k^0{}_{rw}\cdot S_e{}^{\frac{1}{2}}\cdot(1-(1-S_e{}^{1/m})^m)^2$; $m=1-1/n$; + Killough–Land hysteresis	van Genuchten (1980), Soil Sci. Soc. Am. J. 44:892; Mualem (1976), WRR 12:513; Krevor et al. (2012), WRR 48:W02514; Land (1968), Trans. AIME 243:149; Killough (1976), SPE J. 16(1):37
Fracture Pressure	$P_f = [\nu/(1-\nu)](\sigma_v-P_h) + P_h + T_0$; minimum horizontal stress criterion	Hubbert & Willis (1957), Trans. AIME 210:153–166
Mohr-Coulomb	$\tau = c + \sigma_n\tan(\phi)$; $SF = \tau_{strength}/\tau_{drive}$; failure when $SF < 1$	Jaeger, Cook & Zimmerman (2007), Fundamentals of Rock Mechanics, 4th ed.
MAIP	$MAIP = 0.9 \cdot P_f - P_{res}$; maximum allowable injection pressure	Bachu et al. (2007), Int. J. GHG Control 1(4):374; EPA UIC Class VI §146.88
Surface Heave	$\delta \approx (\Delta P \cdot h) / E \cdot (1-\nu^2) \cdot a$; elastic nucleus-of-strain approximation	Teatini et al. (2011), J. Geophys. Res. 116:B08204
Area of Review	$r_{AoR} = \sqrt{(4Kt/\phi\mu c_t)}$; pressure perturbation front ≥ 6.9 kPa threshold	Chabora & Benson (2009), GHGT-9; EPA Class VI UIC Program Guidance

CARBON LENS

About CarbonLens

CarbonLens is a browser-based CO_2 geological storage simulation studio, purpose-built for deep saline aquifer CCS screening. It delivers real-time plume simulation, geomechanical risk assessment, and regulatory permit preparation.

Developed as MSc research at Universiti Teknologi PETRONAS (UTP), Malaysia, in partnership with the CarbonLens product team.

RESEARCH TEAM

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Dr. Okorie Ekwe Agwu — Supervisor, UTP Malaysia
Dr. Muhammed Aslam MD Yusuf — Co-Supervisor, UTP Malaysia

SIMULATION ENGINE

Peng-Robinson EOS · DOE Goodman (2011) · Duan-Sun Solubility · MARS IFT (Olagunju, in prep.) · Nordbotten (2005) Two-Phase AoR · Mohr-Coulomb Geomechanics · van Genuchten–Mualem kr · Killough–Land Hysteresis

IMPORTANT DISCLAIMER

This document is generated by CarbonLens Simulation Studio using scientific correlations and statistical methods. All capacity estimates are preliminary screening values. **This document is NOT a formal regulatory permit application** and has not been reviewed by a competent authority. Independent review by a qualified petroleum or environmental engineer is required before submission to any regulatory body. CarbonLens accepts no liability for decisions made based solely on this screening output.